

Setup Guide

Prerequisites

- Windows 10/11
- PowerShell
- Docker Desktop
- AWS CLI
- Terraform
- kubectl
- Helm

Steps

1. **Provision EKS Cluster** “powershell terraform init terraform apply aws eks update-kubeconfig –region ap-south-1 –name multicloud-eks
2. Build & Push Images docker build -t multicloud-frontend . docker tag multicloud-frontend .dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend:v3 docker push .dkr.ecr.ap-south-1.amazonaws.com/multicloud-frontend:v3
3. Deploy to Kubernetes kubectl apply -f k8s/frontend-deployment.yaml kubectl apply -f k8s/frontend-service.yaml kubectl apply -f k8s/backend-deployment.yaml kubectl apply -f k8s/backend-service.yaml
4. Verify

kubectl get pods kubectl get svc

MONITORING.md

“markdown # Monitoring Setup

Prometheus

“powershell helm repo add prometheus-community https://prometheus-community.github.io/helm-charts helm install prometheus prometheus-community/prometheus

Grafana

helm repo add grafana https://grafana.github.io/helm-charts helm install grafana grafana/grafana

Dashboards CPU usage

Memory usage

Pod health

API latency

Access Grafana via the LoadBalancer external IP.

AI.md

“‘markdown # AI Anomaly Detection

Script Example

```
“‘python import pandas as pd from sklearn.ensemble import IsolationForest
data = pd.DataFrame({"cpu": [20, 22, 19, 95, 21, 20]}) model = IsolationFor-
est(contamination=0.1) model.fit(data) print("Anomaly detection results:",
model.predict(data))
```

Integration Connect to Prometheus metrics

Detect anomalies in CPU/memory

Trigger alerts or auto-scaling

FUTURE.md

“‘markdown # Future Roadmap

Multi-Cloud Expansion

- Azure AKS
- GCP GKE
- Hybrid deployments

Mobile Applications

- **Android (React Native / Kotlin)**
- **iOS (React Native / Swift)**
- Consume backend APIs
- Provide dashboards for different sectors

Sector Use Cases

- Hospital → Patient monitoring dashboards
- BFSI → Transaction anomaly detection
- Travel → Booking and cluster health visualization
- Retail → Inventory and sales monitoring

CI/CD

- GitHub Actions
- Automated builds and deployments

AI Anomaly Detection

Overview

This module adds intelligence to the CloudOps platform by detecting unusual behavior in system metrics (CPU, memory, latency). It helps prevent outages and supports predictive scaling.

Example Script

```
“python import pandas as pd from sklearn.ensemble import IsolationForest
```

Sample CPU usage data

```
data = pd.DataFrame({"cpu": [20, 22, 19, 95, 21, 20]})
```

Train anomaly detection model

```
model = IsolationForest(contamination=0.1) model.fit(data)
```

Predict anomalies (-1 = anomaly, 1 = normal)

```
print("Anomaly detection results:", model.predict(data))
```

Architecture

Current Achievements

- AWS ECR → Docker images
- AWS EKS → Kubernetes cluster
- Terraform → Infrastructure provisioning
- Services → NodePort/LoadBalancer
- Scripts → Secret rotation, AI anomaly detection

Diagram

AWS Cloud	ECR (Docker Registry)	multicloud-frontend:v3
multicloud-backend:latest	EKS Cluster	Frontend Deployment
end Deployment	Services (LoadBalancer)	Monitoring Prometheus

Grafana Scripts Secret Rotation Cert Rotation AI Anomaly
Detection

Future Expansion

- Multi-cloud (Azure, GCP)
 - Mobile apps (Android/iOS)
 - Sector-specific dashboards
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FUTURE.md

“‘markdown # Future Roadmap

Multi-Cloud Expansion

- Extend deployments to **Azure AKS** and **GCP GKE**.
- Hybrid cloud support for resilience and cost optimization.
- Unified Terraform modules for all providers.

Mobile Applications

- **Android (React Native / Kotlin)** and **iOS (React Native / Swift)**.
- Consume backend APIs for dashboards and alerts.
- Provide native mobile experience for operators and end-users.

Sector Use Cases

- **Hospital** → Patient monitoring dashboards, appointment tracking.
- **BFSI** → Transaction anomaly detection, fraud alerts.
- **Travel** → Booking status dashboards, cluster health visualization.
- **Retail** → Inventory and sales monitoring, demand forecasting.

CI/CD

- GitHub Actions pipelines for automated builds and deployments.
- Continuous testing and security scans.
- Multi-environment deployments (dev, staging, prod).

Security Automation

- Automated **secret rotation**.
- Certificate renewal scripts.
- ECR cleanup and image lifecycle policies.

Resume Impact

This roadmap demonstrates the ability to: - Build a **multi-cloud, full-stack platform**. - Extend into **mobile applications** for cross-sector use. - Integrate **AI and monitoring** for enterprise reliability.